

Faculty Of Engineering and Technology

Electrical and Computer Engineering Department

ENEE 4113, Communication Laboratory

Normal Amplitude Modulation and Demodulation

Pre-Lab Experiment #1

Prepared by:

Shadi Ataya, 1230159.

Instructor: Dr. Nofal Nofal

Teacher Assistant: Eng. Hazem Awaysa

Section: 5.

Date: 12|Oct|2025

Table of contents

Table of figures 4

Message Signal 5

Block Diagram: 5

Parameters: 5

Message signal in time domain: 6

Message signal in frequency domain 6

Carrier Signal 7

Block Diagram: 7

Parameters: 7

Carrier signal in time domain: 8

Carrier signal in frequency domain 8

Amplitude Modulation 9

Block Diagram: 9

1. **Under modulation ($\mu < 1$):** 9

Time Domain: 9

Frequency Domain: 10

2. **Normal modulation ($\mu = 1$):** 10

Time Domain: 10

Frequency Domain: 11

3. **Overmodulation ($\mu > 1$):** 11

Time Domain: 11

Frequency Domain: 12

Amplitude Demodulation 12

First method: Coherent Am demodulation. 12

Block Diagram: 12

1. **Demodulation when μ <1** 13

Time Domain: 13

Frequency Domain: 13

2. **Demodulation when μ =1** 14

Time Domain: 14

Frequency Domain: 14
3. Demodulation when $\mu > 1$ 15
Time Domain: 15
Frequency Domain: 15
Second method: Envelop Detector Demodulation. 16
Block Diagram: 16
1. Demodulation when μ <1 16
Time Domain: 16
Frequency Domain: 17
2. Demodulation when μ =1 17
Time Domain: 17
Frequency Domain: 17
3. Demodulation when μ >1 18
Time Domain: 18
Frequency Domain: 18

Table of figures

Figure 1 Message signal block diagram 5
Figure 2 Message signal parameters 5
Figure 3 Message signal in time domain 6
Figure 4 Message signal in frequency domain 6
Figure 5 Carrier signal block diagram 7
Figure 6 Carrier signal parameters 7
Figure 7 Carrier signal in time domain 8
Figure 8 Carrier signal in frequency domain 8
Figure 9 Amplitude modulation diagram 9
Figure 10 Under modulation $\mu < 1$ in time domain 9
Figure 11 Under modulation $\mu < 1$ in frequency domain 10
Figure 12 Normal modulation $\mu = 1$ in time domain 10
Figure 13 Normal modulation $\mu = 1$ in frequency domain 11
Figure 14 Overmodulation $\mu > 1$ in time domain 11
Figure 15 Overmodulation $\mu > 1$ in frequency domain 12
Figure 16 Coherent demodulation block diagram 12
Figure 17 Demodulation using coherent in time domain when $\mu <1$ 13
Figure 18 Demodulation using coherent in frequency domain when μ <1 13
Figure 19 Demodulation using coherent in time domain when μ =1 14
Figure 20 Demodulation using coherent in frequency domain when μ =1 14
Figure 21 Demodulation using coherent in time domain when μ >1 15
Figure 22 Demodulation using coherent in frequency domain when μ >1 15
Figure 23 Envelop detector demodulation block diagram 16
Figure 24 Demodulation using Envelope in time domain when μ <1 16
Figure 25 Demodulation using Envelope in frequency domain when μ <1 17
Figure 26 Demodulation using Envelope in time domain when μ =1 17
Figure 27 Demodulation using Envelope in frequency domain when μ =1 17
Figure 28 Demodulation using Envelope in time domain when μ >1 18
Figure 29 Demodulation using Envelope in frequency domain when μ >1 18

Message Signal

m(t) = 0.85 cos (2π (1000) t).

Block Diagram:

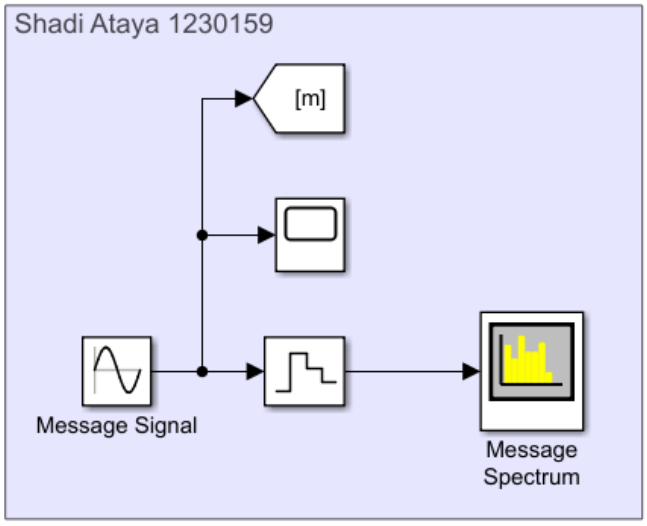


Figure 1 Message signal block diagram

Parameters:

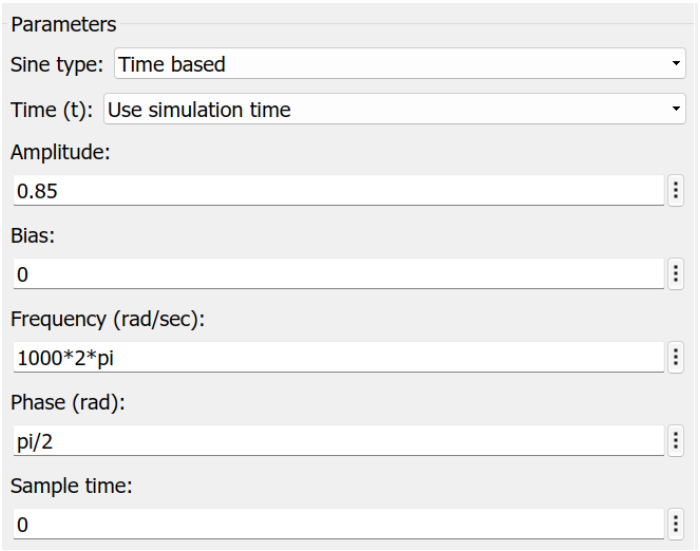


Figure 2 Message signal parameters

Message signal in time domain:

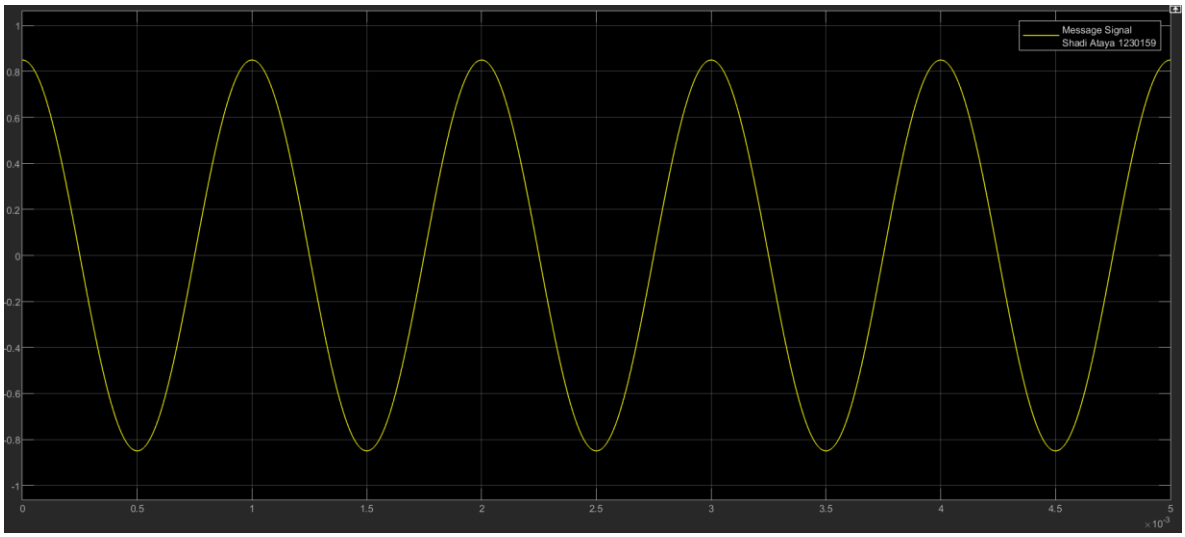


Figure 3 Message signal in time domain

Message signal in frequency domain

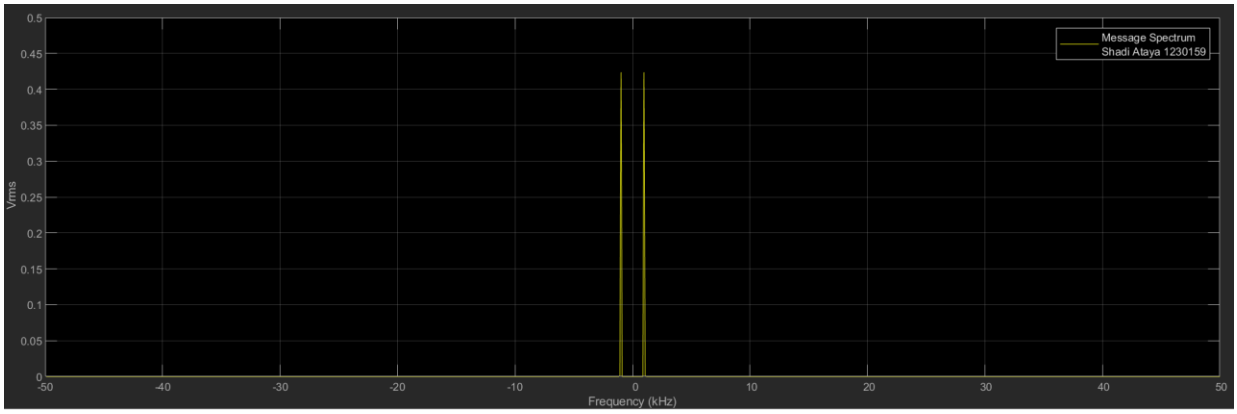


Figure 4 Message signal in frequency domain

Carrier Signal

c(t) = 1cos (2π (15k) t).

Block Diagram:

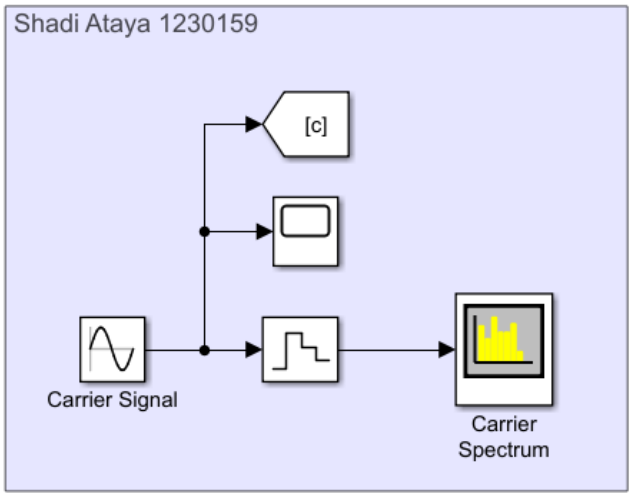


Figure 5 Carrier signal block diagram

Parameters:

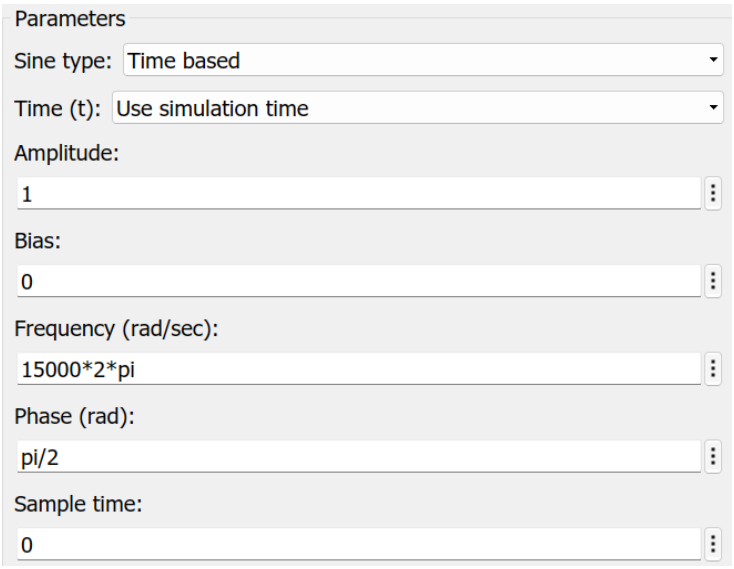


Figure 6 Carrier signal parameters

Carrier signal in time domain:

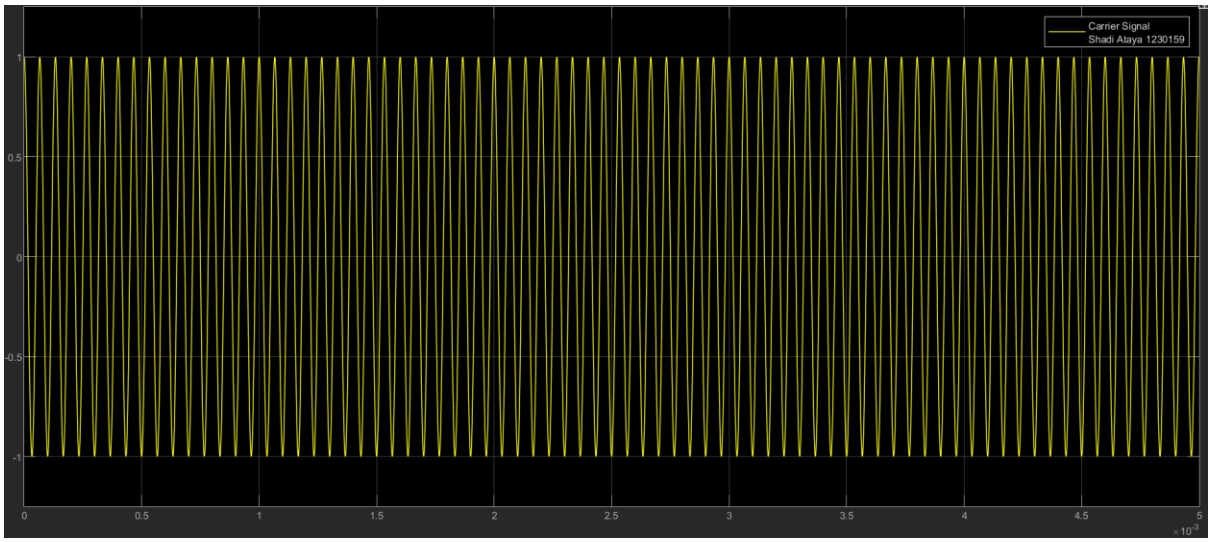


Figure 7 Carrier signal in time domain

Carrier signal in frequency domain

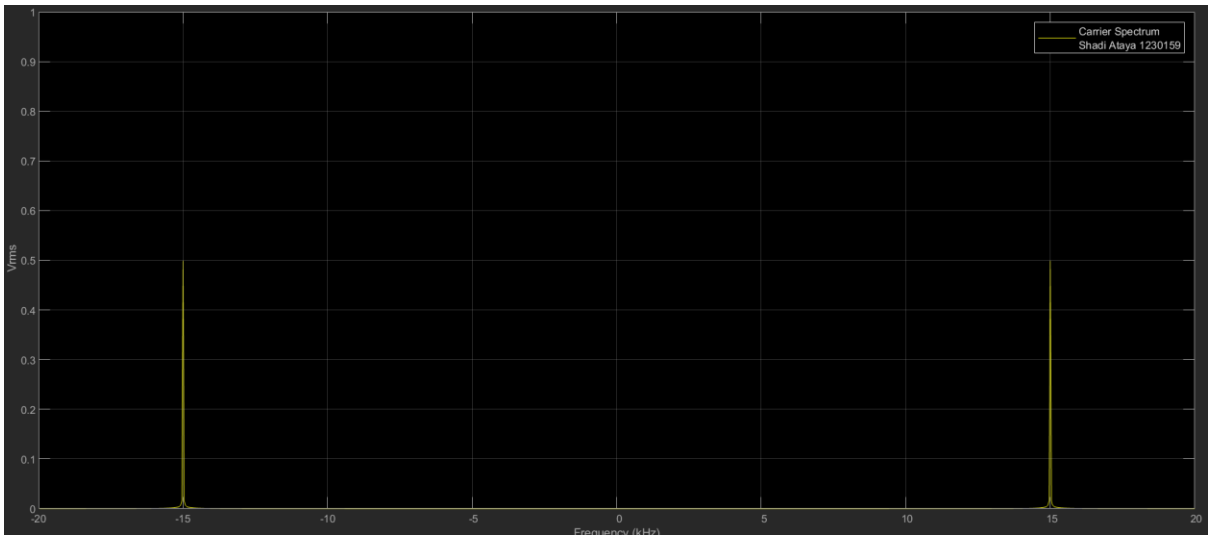


Figure 8 Carrier signal in frequency domain

Amplitude Modulation

Block Diagram:

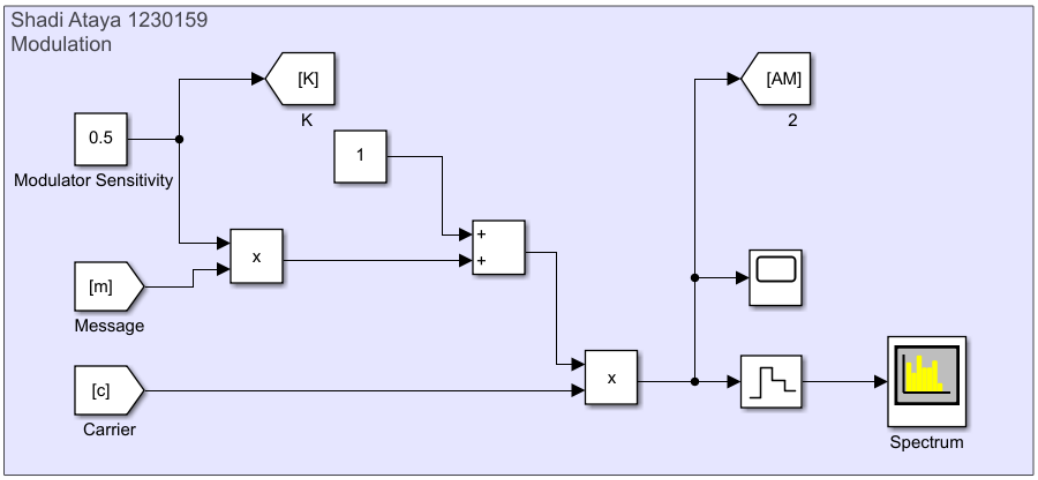


Figure 9 Amplitude modulation diagram

1. Under modulation ($\mu < 1$):

Using K = 0.5 so $\mu = 0.425$

Time Domain:

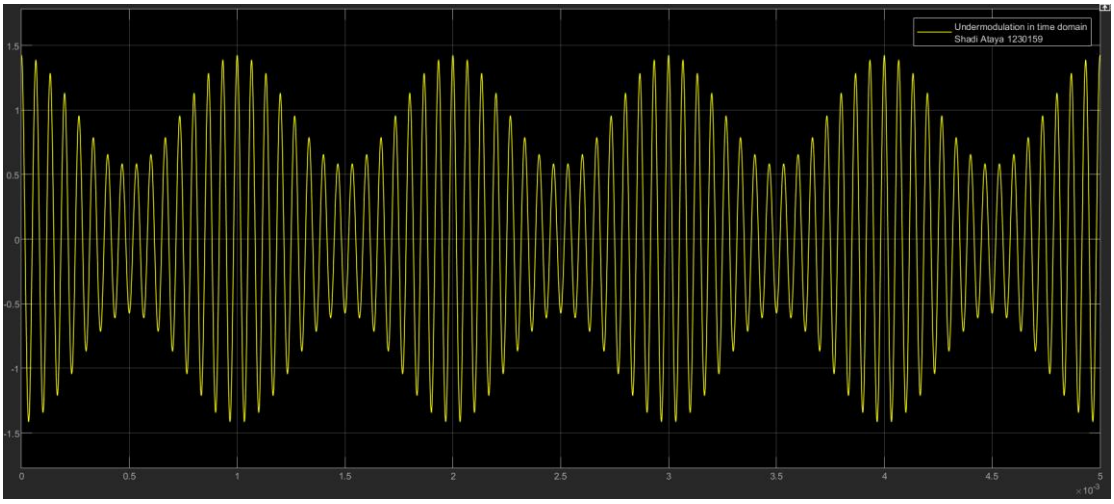


Figure 10 Under modulation $\mu < 1$ in time domain

Frequency Domain:

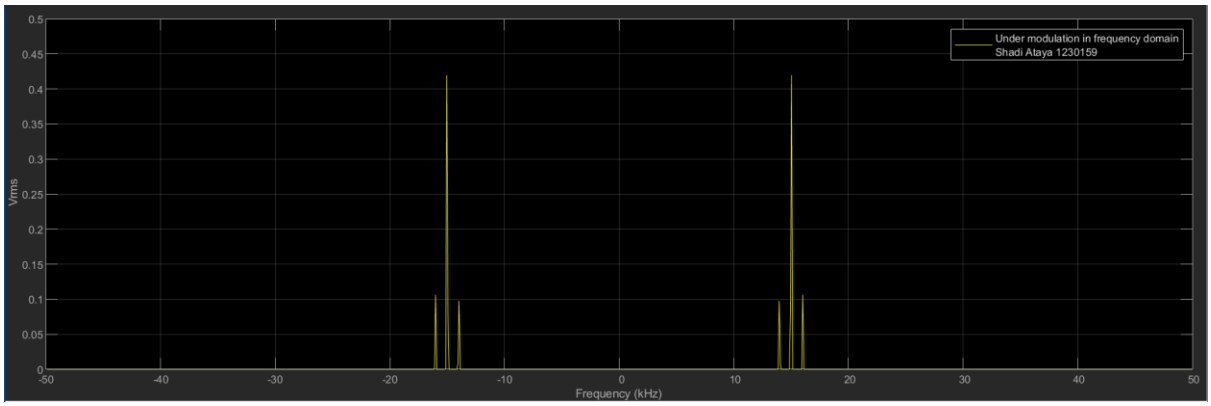


Figure 11 Under modulation μ < 1 in frequency domain

2. Normal modulation ($\mu = 1$):

Using K = 1/0.85 so $\mu = 1$

Time Domain:

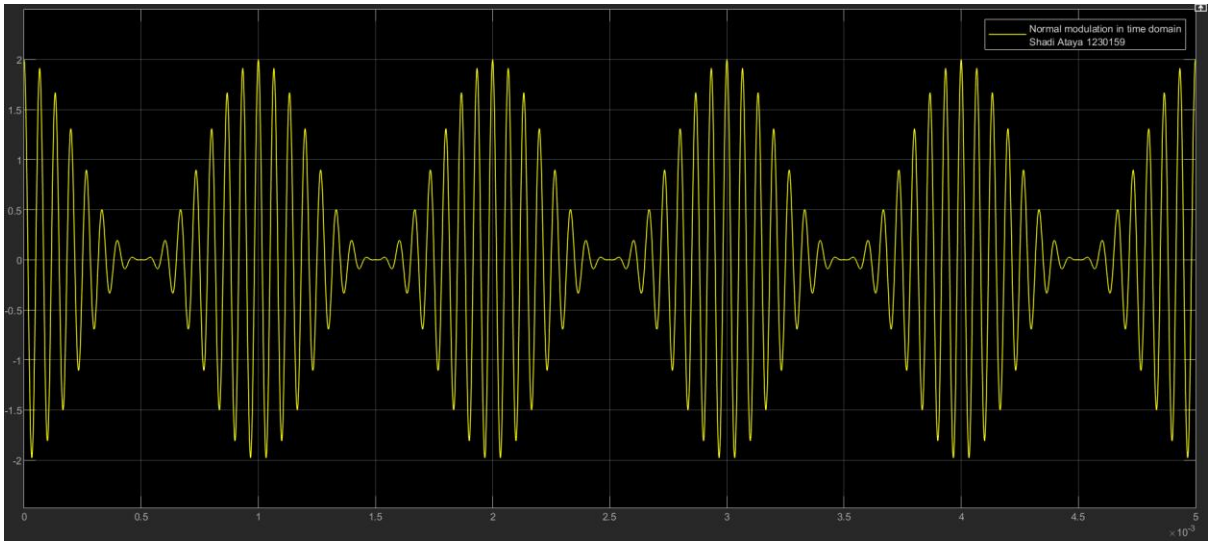


Figure 12 Normal modulation μ = 1 in time domain

Frequency Domain:

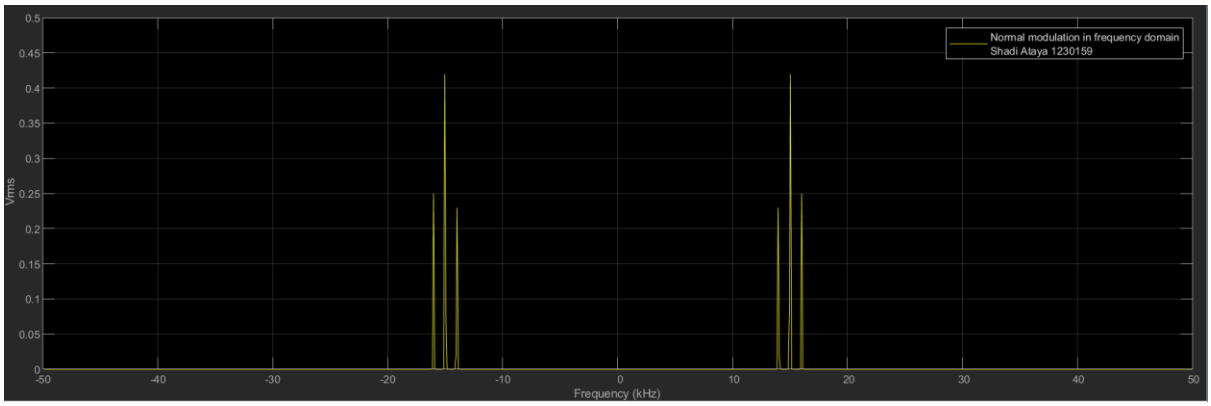


Figure 13 Normal modulation μ = 1 in frequency domain

3. Overmodulation ($\mu > 1$):

Using K = 3 so $\mu = 2.55$

Time Domain:

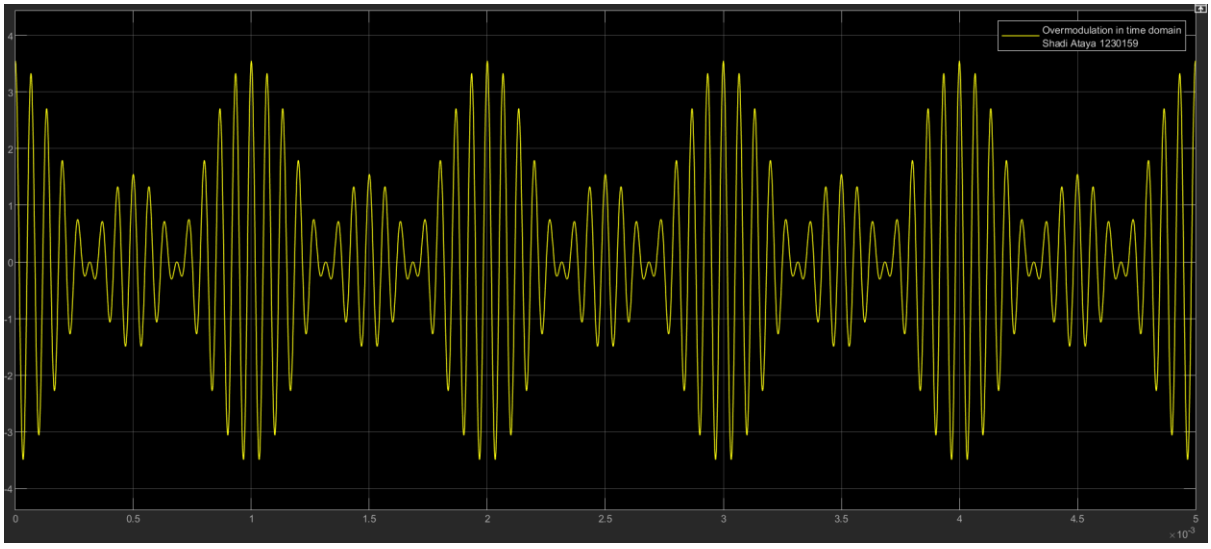


Figure 14 Overmodulation μ > 1 in time domain

Frequency Domain:

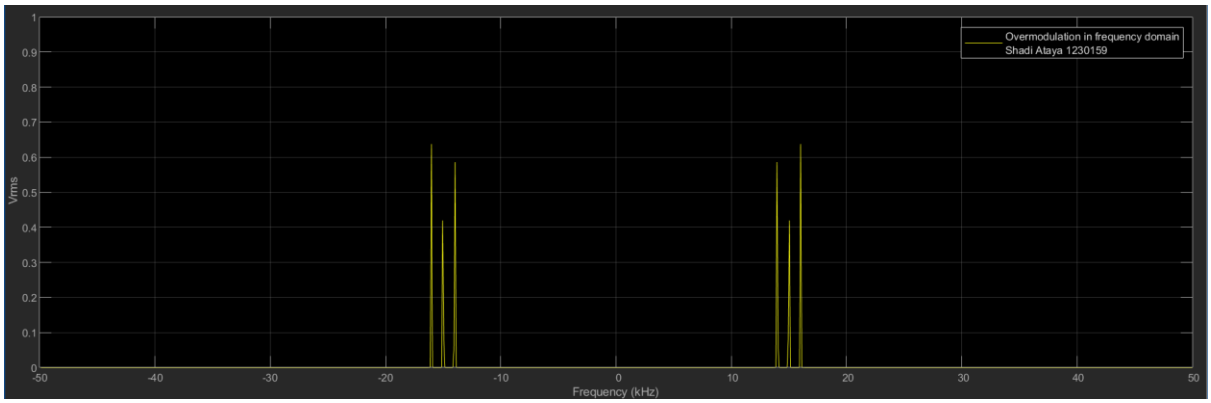


Figure 15 Overmodulation μ > 1 in frequency domain

Amplitude Demodulation

First method: Coherent Am demodulation.

Block Diagram:

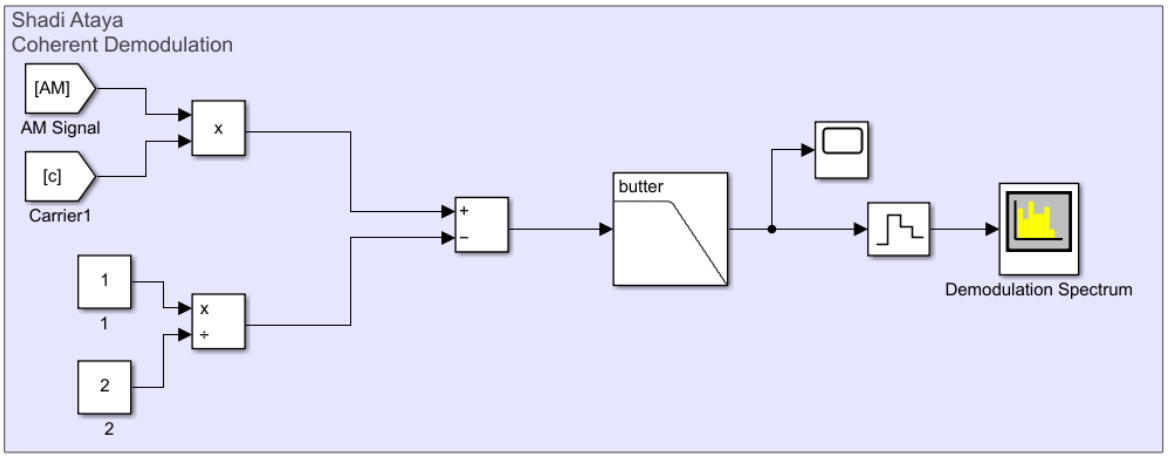


Figure 16 Coherent demodulation block diagram

1. Demodulation when μ <1

Time Domain:

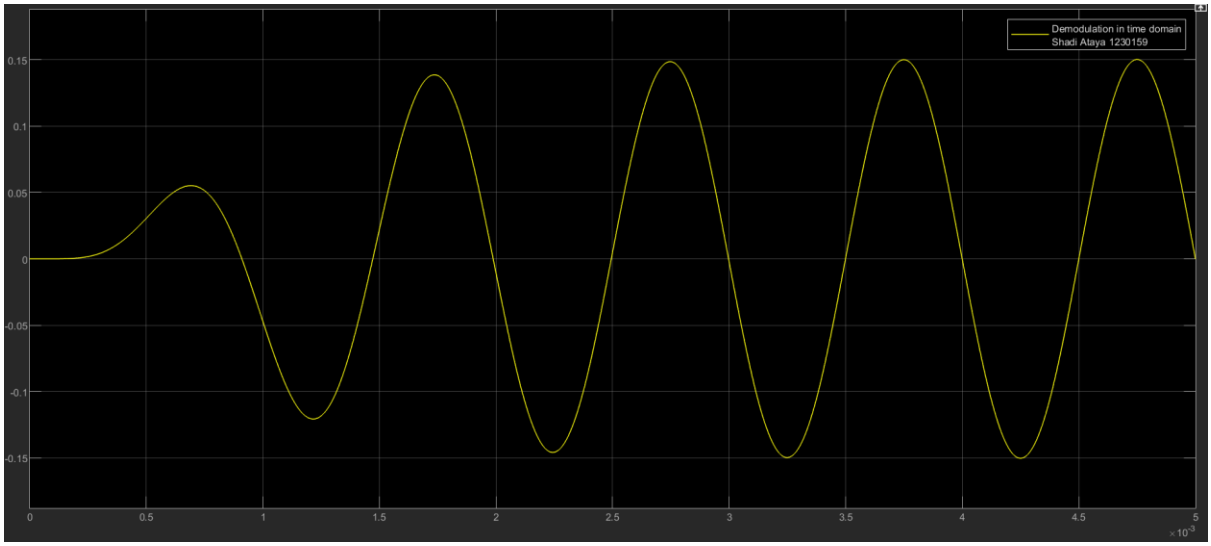


Figure 17 Demodulation using coherent in time domain when μ <1

Frequency Domain:

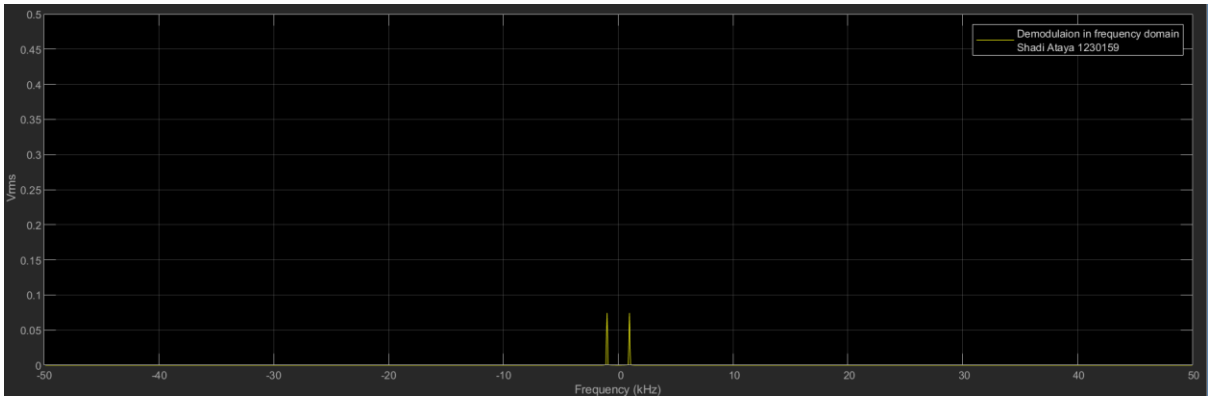


Figure 18 Demodulation using coherent in frequency domain when μ <1

2. Demodulation when μ =1

Time Domain:

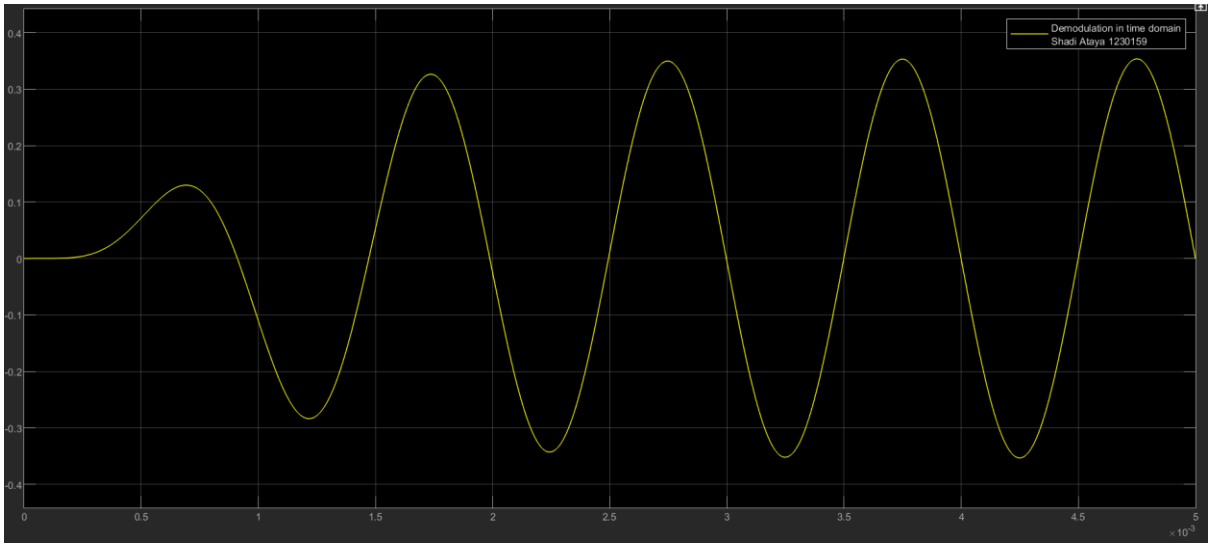


Figure 19 Demodulation using coherent in time domain when μ =1

Frequency Domain:

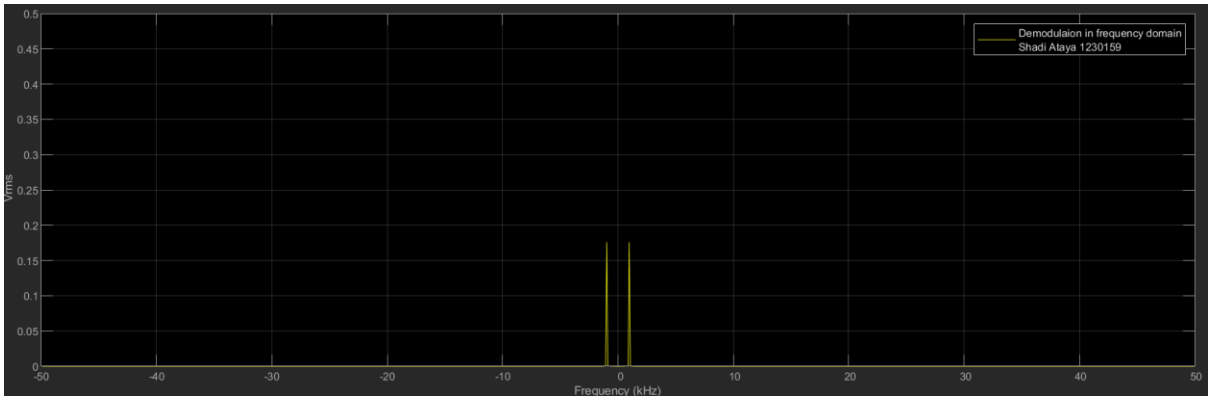


Figure 20 Demodulation using coherent in frequency domain when μ =1

3. Demodulation when $\mu > 1$

Time Domain:

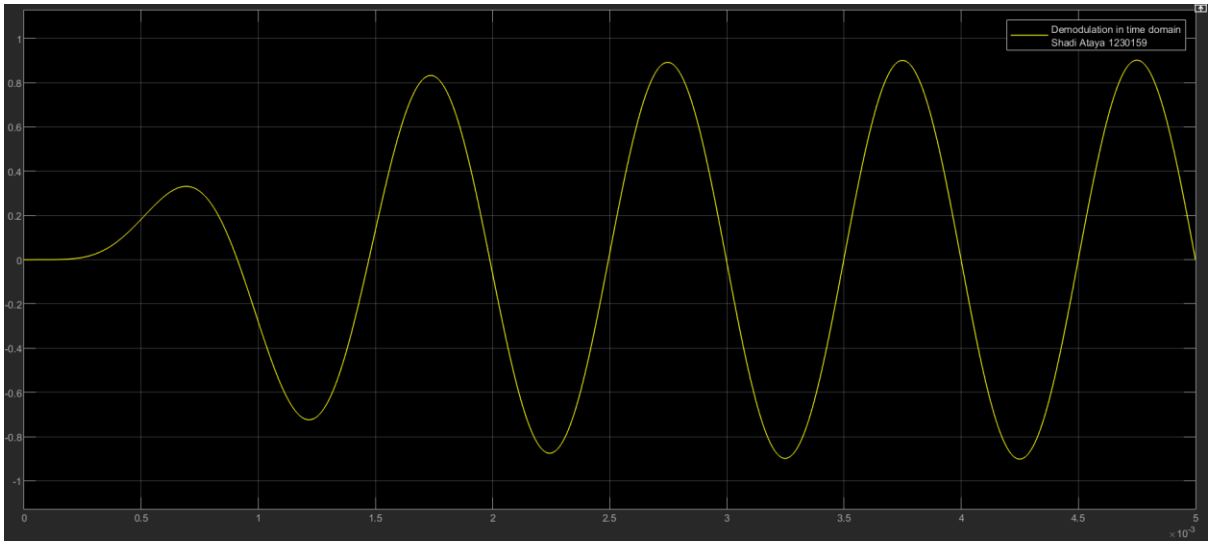


Figure 21 Demodulation using coherent in time domain when μ >1

Frequency Domain:

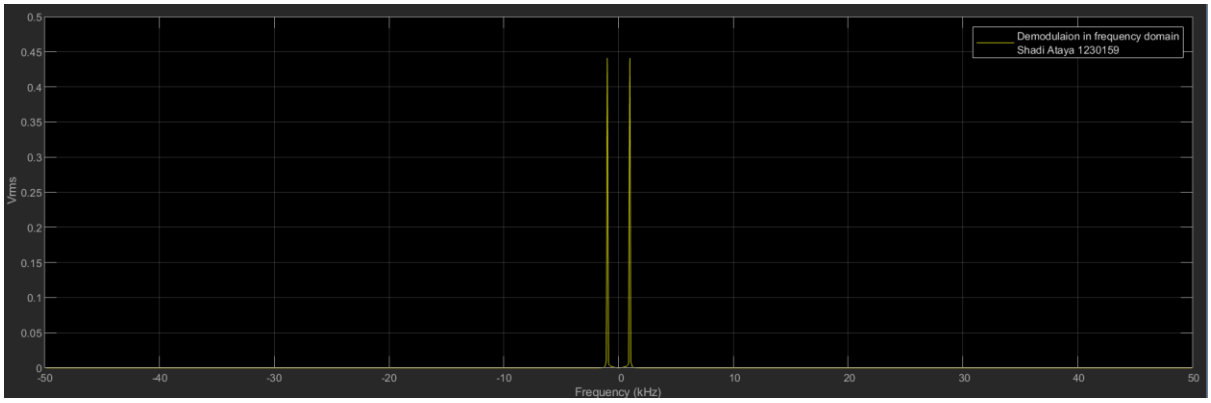


Figure 22 Demodulation using coherent in frequency domain when μ >1

Second method: Envelop Detector Demodulation.

Block Diagram:

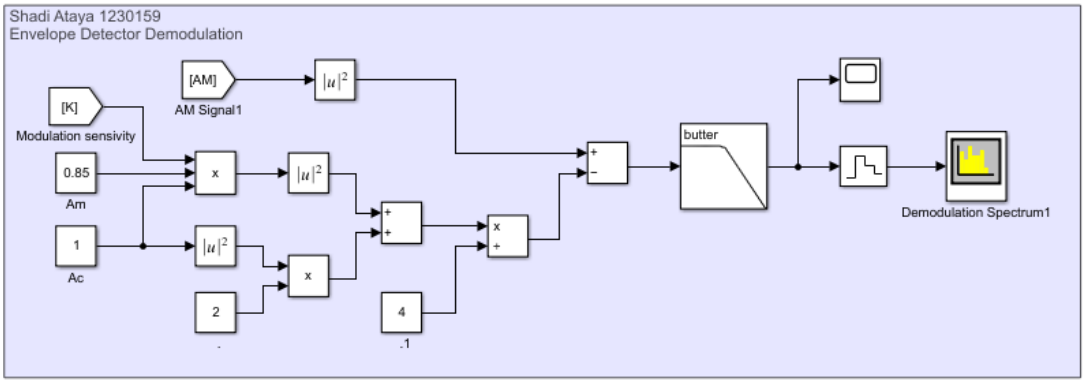


Figure 23 Envelop detector demodulation block diagram

1. Demodulation when μ <1

Time Domain:

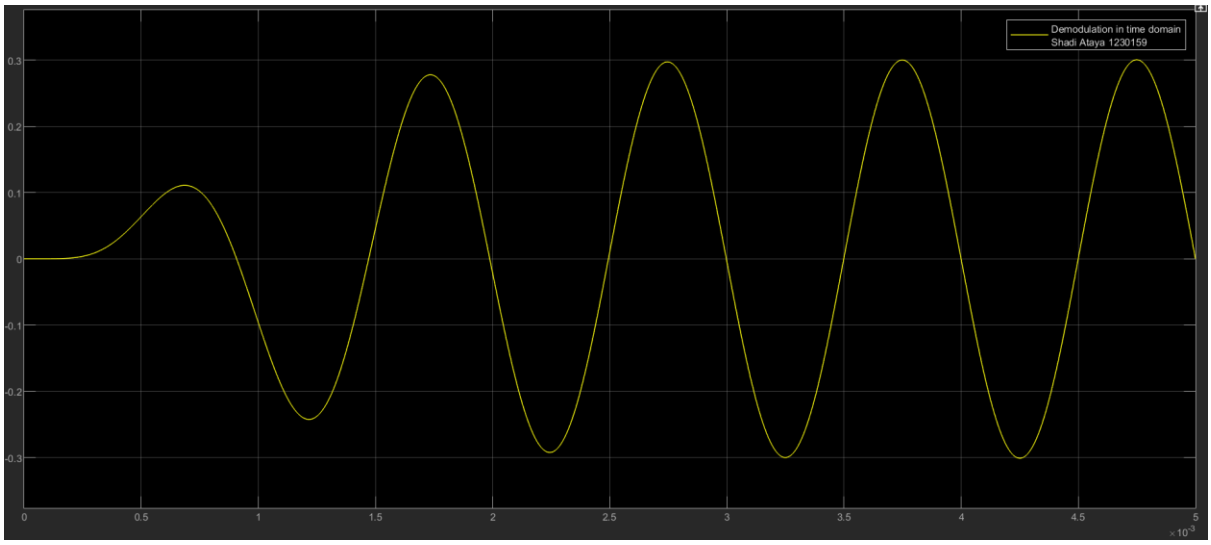


Figure 24 Demodulation using Envelope in time domain when μ <1

Frequency Domain:

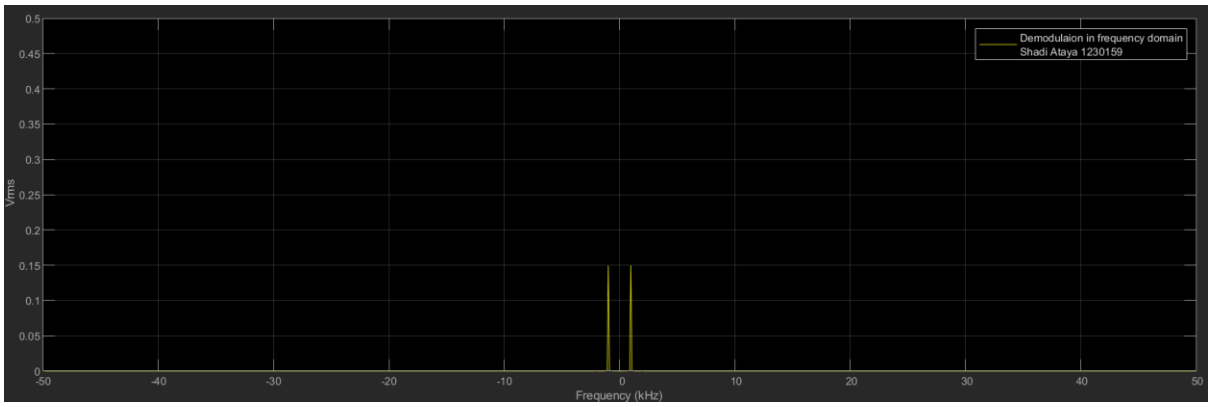


Figure 25 Demodulation using Envelope in frequency domain when μ <1

2. Demodulation when μ =1

Time Domain:

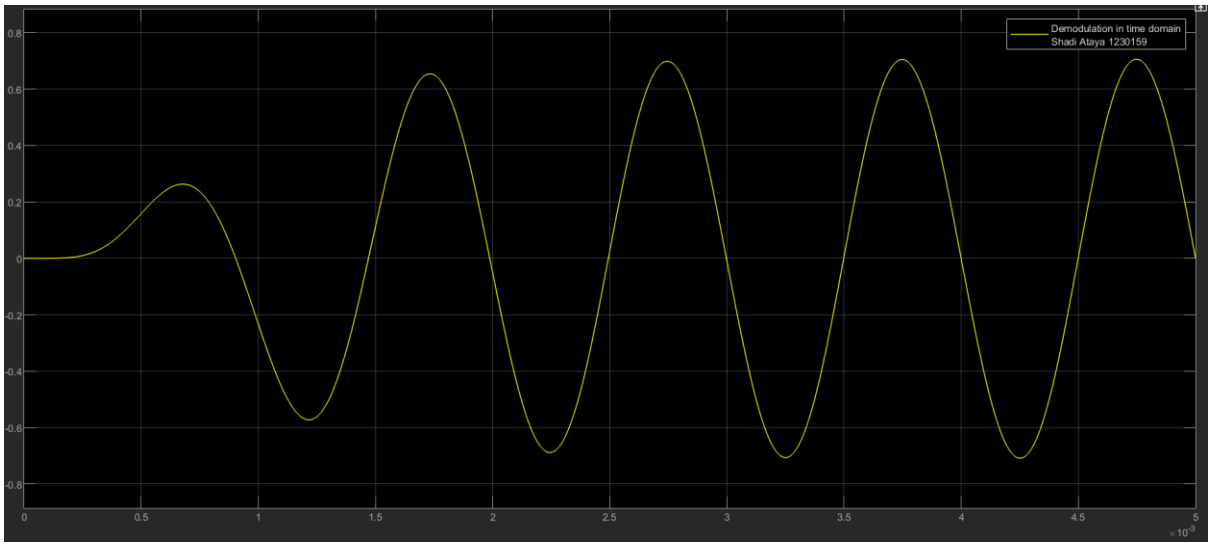


Figure 26 Demodulation using Envelope in time domain when μ =1

Frequency Domain:

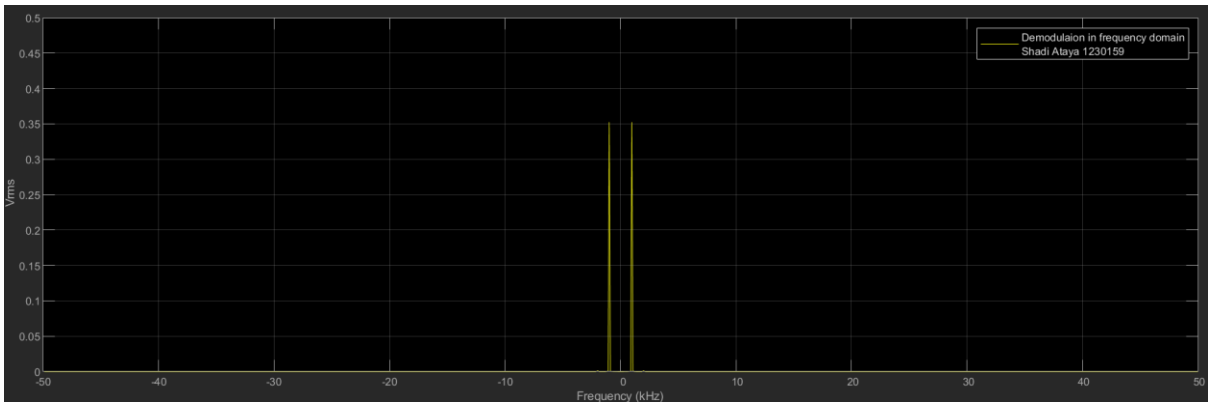


Figure 27 Demodulation using Envelope in frequency domain when μ =1

3. Demodulation when μ >1

Time Domain:

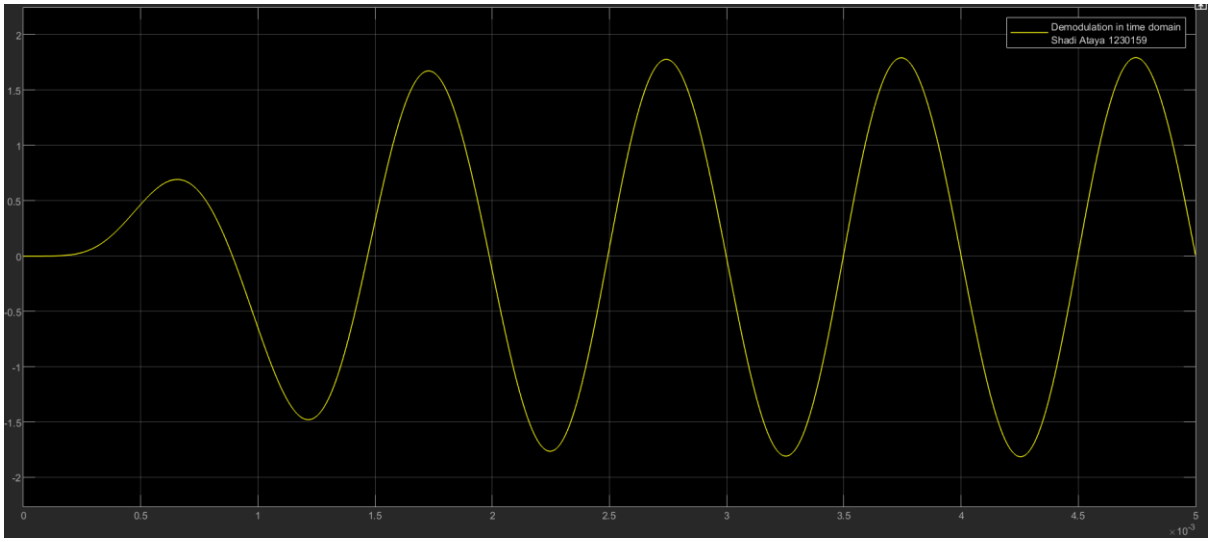


Figure 28 Demodulation using Envelope in time domain when μ >1

Frequency Domain:

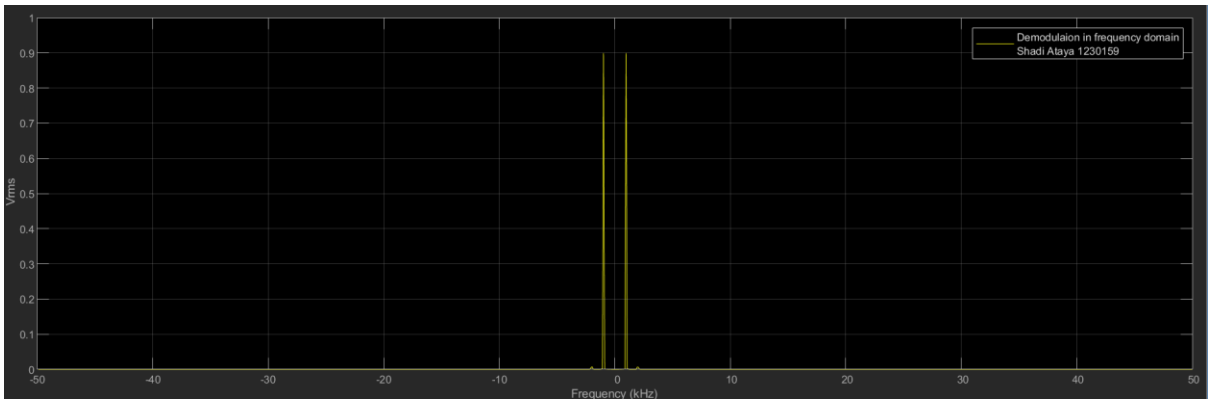


Figure 29 Demodulation using Envelope in frequency domain when μ >1